Project Report

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Problem Statement: This project aims to develop a deep learning-based solution that can classify aerial images into two categories — Bird or Drone

Created a virtual environment inside the folder bird-drone-classifier

Project WorkFlow:

1. In the workflow, the Jupyter notebook used for exploratory data analysis (EDA), sample visualization, inspection of labels, and quick experimentation with preprocessing or loading methods. With the file naming scheme, each label .txt file matches one image .jpg file by name done.
2. Then ran train.py file, (**python train.py in terminal)** , it loaded the image/label loading logic and monitored the training progress(accuracy, loss etc.) in console. After training the trained model stored in bird_drone_best.pth
3. Then ran the evaluate.py, (**python evaluate.py in terminal**). It helped in assessing the performance of the trained model using validation set, preprocess validation images and labels, displayed confusion matrix and classification report and distinguished between bird and drone.
4. Then ran yolo.py file to include object detection using YOLOv8. This allowed models to not only classify images as bird or drone but also localize birds and drones with image.

   Folder runs/detec/train is created containing model weights, training plots(etc).

5. Then ran yolotest.py file that gave metrics on console (map, precision, recall) on unseen data before deployment.
6. Then started the deployment, ran streamlit app.py and the browser showed the Bird & Drone Detection (YOLOv8) app. Browsed files, uploaded file and the app correctly distinguished image as drone or bird and localized image.